

Numerical Assessment of Microclimate in a Plant Factory Using CFD

EBS 270 Final Presentation

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Core question: How do airflow, heat transfer, and moisture transport shape the lettuce-surface microclimate in an indoor plant factory chamber?

Motivation and Research Gap

Why local microclimate matters

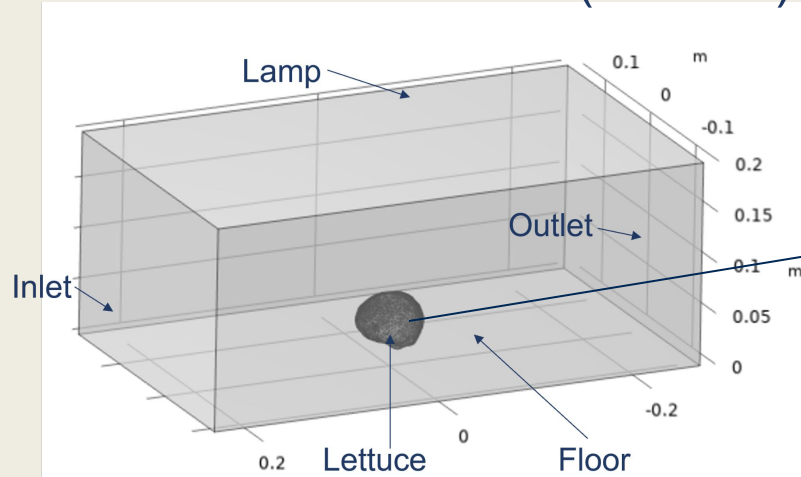
- Lettuce response is driven by leaf-surface temperature, humidity, and air speed.
- Bulk chamber setpoints may hide local gradients near irregular leaves.
- CFD can resolve spatial fields that are difficult to measure experimentally.



System Geometry and Physical Domains

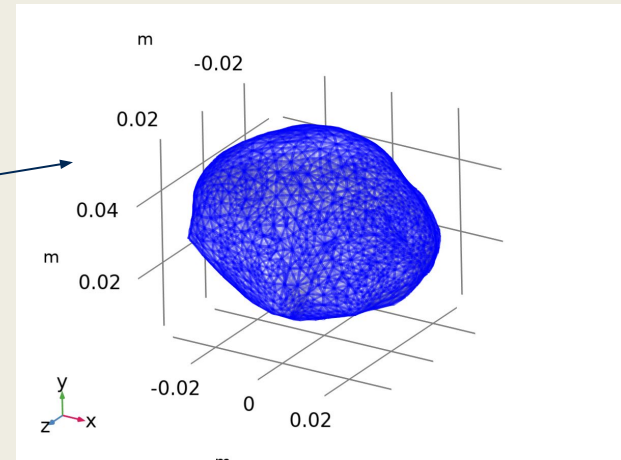
Air domain:

0.3 m × 0.3 m × 0.7 m (H×W×L)



Lettuce domain:

Diameter = 0.065m



Problem Statement and Objectives

Problem statement:

The lettuce microclimate is governed by the coupled transport of momentum, heat, and water vapor around an irregular plant geometry under forced ventilation and LED heating.

Objectives:

- Develop a 3-D CFD model of the microclimate around lettuce
- Compare analytical and numerical analysis of lettuce temperature
- Compare the effect the different control scheme of inlet temperature
- Investigate the effects of inlet velocity and inlet temperature on microclimate conditions

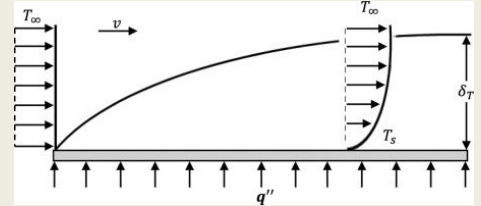
Model Assumptions and Simplifications

Lettuce domain

- Treated as a homogeneous biological solid
- No internal airflow or internal liquid-water transport
- Heat transfer by transient conduction

Thermal boundaries

- LED modeled as a constant-temperature boundary
- Radiative transfer not solved explicitly
- Inlet flow prescribed as uniform velocity and temperature

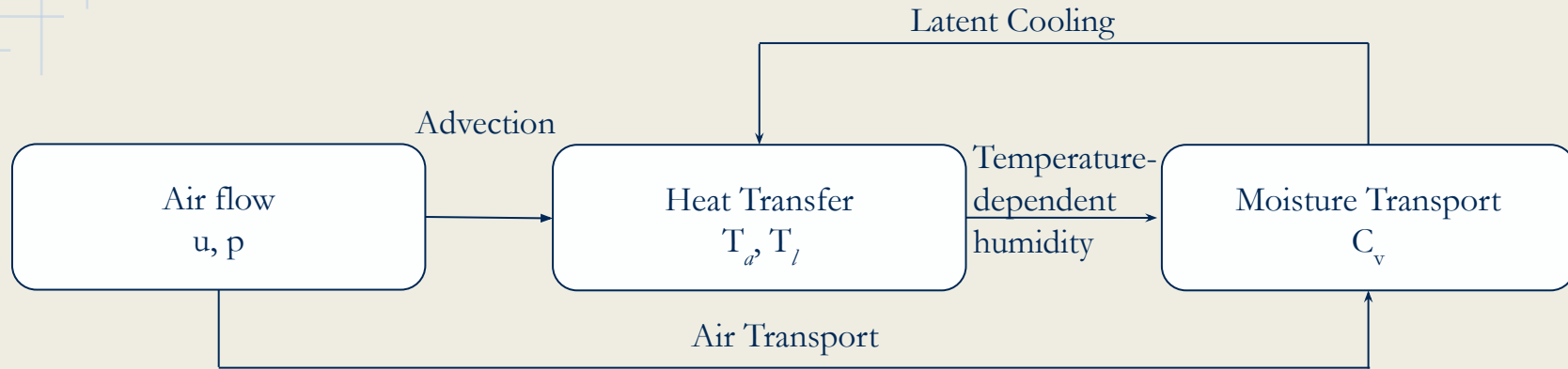


Moisture domain

- Water vapor transported only in air
- Lettuce surface acts as transpiration-flux boundary



Governing Physics: Coupled Transport System



Domain	Included Physics
Air	Flow, heat transfer, vapor transport
Lettuce	Internal heat conduction
Lettuce–air surface	Convection, transpiration, latent cooling
LED boundary	Thermal input from lighting
Chamber walls	No-slip, heat exchange, vapor no-flux

Air domain

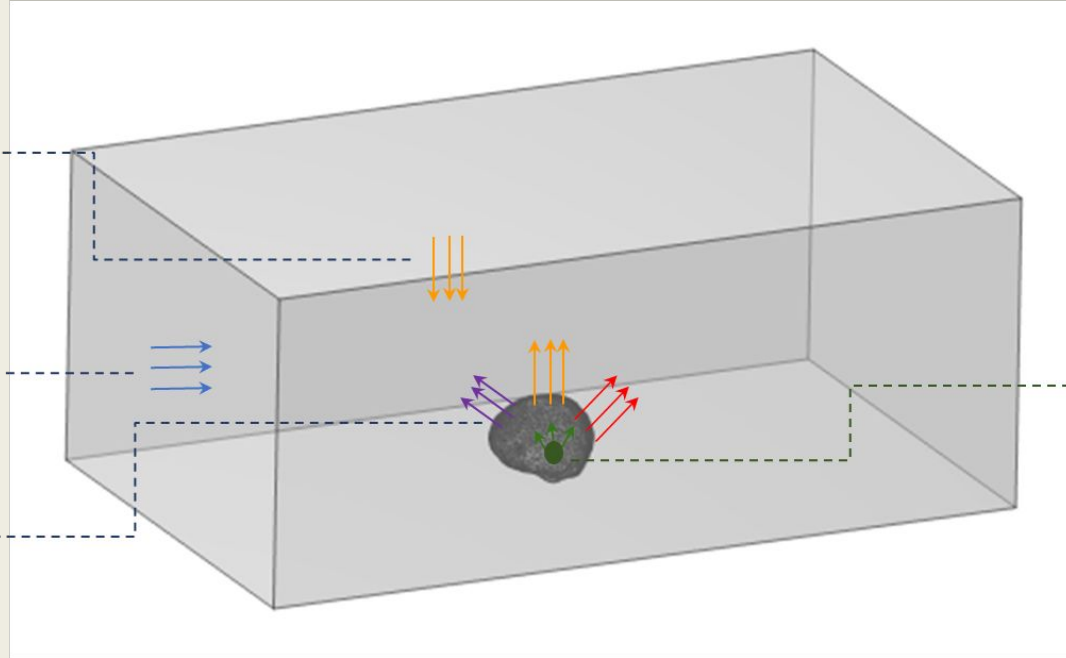
lettuce domain

Heat
balance

Momentum
balance

Mass
balance

Heat
balance



→ Air flow

→ Transpiration heat

→ Water vapor

→ Convection

→ Conduction

● Metabolic heat

Governing equations overview

Momentum balance in the Air Domain

Physics	Governing Equation	Variable	Initial Condition	Boundary Condition
Momentum balance	$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = -\nabla p + \mu \nabla^2 \mathbf{u} + \rho \mathbf{g}$ $\nabla \cdot \mathbf{u} = 0$	Air velocity ($\mathbf{u}$) Pressure (p)	$\mathbf{u}(x, y, z, 0) = 0$ $p(x, y, z, 0) = p_{\text{atm}}$	Inlet: $\mathbf{u} = \mathbf{u}_{in}$ Walls, LED, lettuce surface (No-slip Condition): $\mathbf{u} = 0$ Outlet: $p = p_{out} = 0\text{Pa}$

Energy balance in the Air domain

Physics	Governing Equation	Variable	Initial Condition	Boundary Condition
Energy balance in air domain	$\rho_a C_{p,a} \left(\frac{\partial T_a}{\partial t} + \mathbf{u} \cdot \nabla T_a \right) = \nabla \cdot (k_a \nabla T_a)$	Air temperature, T_a	$T_a(x, y, z, 0) = T_{a,0}$	Inlet: $T_a = T_{\text{in}}$ Outlet: $-\mathbf{n} \cdot (k_a \nabla T_a) = 0$ LED surface: $T_a = T_{\text{LED}}$ Chamber walls: $-\mathbf{n} \cdot (k_a \nabla T_a) = h_{\text{wall}}(T_a - T_{\infty})$ Lettuce--air interface: heat exchange is coupled with the lettuce surface through convection and transpiration cooling.

Energy balance in the Air domain

Physics	Governing Equation	Variable	Initial Condition	Boundary Condition
Energy balance in lettuce domain	$\rho_l C_{p,l} \frac{\partial T_l}{\partial t} = \nabla \cdot (k_l \nabla T_l) + Q_l$	Lettuce temperature, T_l Lettuce metabolic heat generation, Q_l	$T_l(x, y, z, 0) = T_{l,0}$	Lettuce--air interface: $-\mathbf{n} \cdot (k_l \nabla T_l) = h_c(T_l - T_a) + \lambda E$ Internal lettuce domain: heat transfer occurs by conduction only. If metabolic heat is neglected: $Q_l = 0$

Mass balance in the Air domain

Physics	Governing Equation	Variable	Initial Condition	Boundary Condition
Water-vapor mass balance in air domain	$\frac{\partial c_v}{\partial t} + \mathbf{u} \cdot \nabla c_v = \nabla \cdot (D_v \nabla c_v)$	Water-vapor concentration, c_v	$c_v(x, y, z, 0) = c_{v,0}$ If initialized from relative humidity: $c_{v,0} = \frac{RH_0}{100} \frac{p_{\text{sat}}(T_{a,0})}{R_v T_{a,0}}$	Inlet: $c_v = c_{v,\text{in}}$ $c_{v,\text{in}} = \frac{RH_{\text{in}}}{100} \frac{p_{\text{sat}}(T_{\text{in}})}{R_v T_{\text{in}}}$ Outlet: $-\mathbf{n} \cdot (D_v \nabla c_v) = 0$ Lettuce surface: $-\mathbf{n} \cdot (D_v \nabla c_v) = E$ Chamber walls and LED: $-\mathbf{n} \cdot (D_v \nabla c_v) = 0$
Relative humidity	$RH = 100 \frac{c_v R_v T_a}{p_{\text{sat}}(T_a)}$	Relative humidity, RH	$RH(x, y, z, 0) = RH_0 = 65\%$	Relative humidity is derived from c_v and T_a , rather than solved as an independent governing equation.

Lettuce surface energy balance

Physics	Governing Equation	Variable	Initial Condition	Boundary Condition
Lettuce surface energy balance	$\lambda E + H = \pi R_{\gamma} + Q_h - G - K_s \Delta T$	Leaf surface temperature, (T_s) Sensible heat flux, (H) Latent heat flux, (λE)	$T_s(t = 0) = T_{s,0}$ Common assumption: $T_{s,0} = T_0$ $H(t = 0) = h_c(T_{s,0} - T_0)$ $E(t = 0) = E_0$	Lettuce surface: $-\mathbf{n} \cdot (k \nabla T) = H + \lambda E$ $H = h_c(T_s - T_a)$ $-\mathbf{n} \cdot (D_v \nabla c_v) = E$

COMSOL model and Simulation setup

▼ Turbulent Flow, Realizable k- ϵ 2 (*spf2*)

▼ Fluid Properties 1 Equation View

- > Initial Values 1
- > Wall 1
- > Gravity 1
- > Symmetry 1
- > Inlet 1
- > Outlet 1
- > Wall 2

Equation View

▼ Heat Transfer in Solids and Fluids (*ht*)

- > Solid 1
- > Fluid 1
- > Initial Values_air
- > Thermal Insulation 1
- > LED
- > Inlet
- > Symmetry 1
- > Outflow 1
- > Initial Values_lettuce
- > Temperature 3
- > Heat Source 1
- Equation View

▼ Moisture Transport in Air (*mt*)

- > Moist Air 1
- > Initial Values 1
- > Insulation 1
- > Moisture Flux 1
- > Inflow 1
- > Outflow 1
- > Symmetry 1
- Equation View

▼ Multiphysics

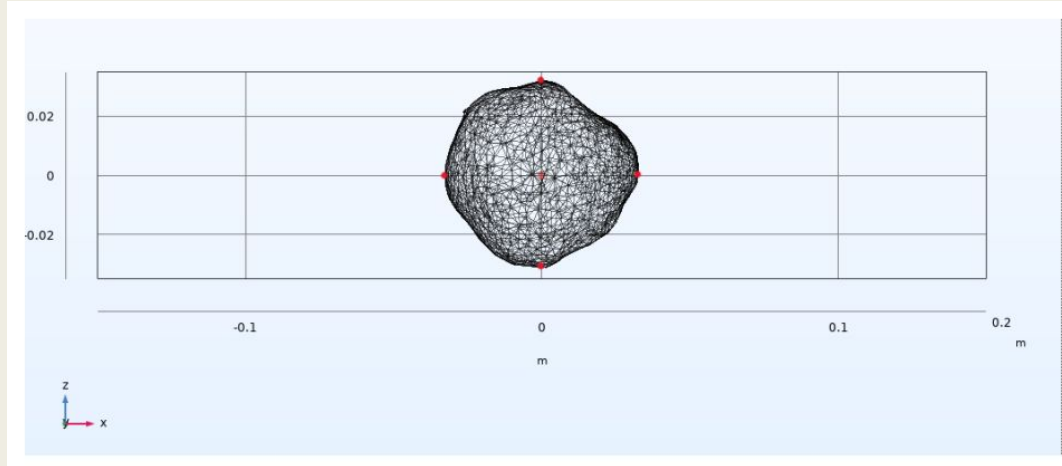
- > Nonisothermal Flow 1 (*nitf1*)
- > Moisture Flow 1 (*mf1*)

Submodel/Study	Input Parameter	Value	Unit
Turbulent flow	Initial velocity	0	m s^{-1}
	Acceleration of gravity	(0, - g_{const} , 0)	m s^{-2}
	Inlet velocity	0.7, 0.8, 0.9, 1.1	m s^{-1}
	Outlet pressure	0	Pa
Heat transfer in solid and fluid	Lettuce initial temperature	297.65	K
	Air initial temperature	298.15	K
	LED temperature	313.15, 343.15, 383.15	K
	Inlet temperature	291.15, 293.15, 295.15	K
	Heat rate of lettuce	PID-controlled setpoint (293.15) 0.004, 0	W
Moisture transport in air	Initial relative humidity	65	%
	Moisture flux	E^*	$\text{kg m}^{-2} \text{s}^{-1}$
	Inlet humidity	65	%
Time-dependent study	Time range	0 : 1 : 180	min

Material Properties Used in the Model

Domain	Property	Symbol	Value
Lettuce	Density	ρ_l	1078 kg m^{-3}
Lettuce	Specific heat	$C_{p,l}$	$3100 \text{ J kg}^{-1} \text{ K}^{-1}$
Lettuce	Thermal conductivity	k_l	$0.55 \text{ W m}^{-1} \text{ K}^{-1}$

Probe Locations for Parametric and Sensitivity Analysis

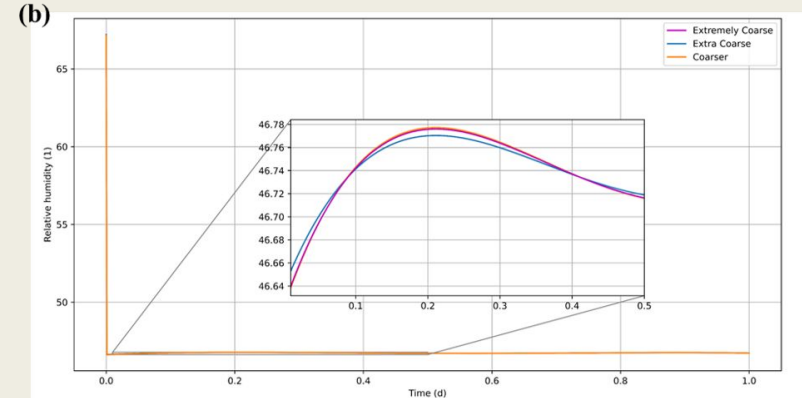
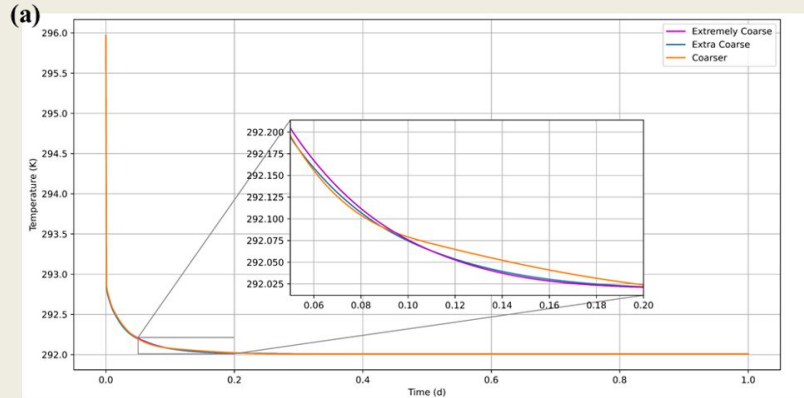


- All reported values were the average values of the 5 probes

Probe	<i>x</i> -coordinate	<i>y</i> -coordinate	<i>z</i> -coordinate
Probe 1	0	0.027	0.032
Probe 2	0	0.027	-0.032
Probe 3	0.033	0.027	0
Probe 4	-0.033	0.027	0
Probe 5	0	0.055	0

Mesh Independence Test

Mesh Case	Average Temp.	Max Temp.	Average RH	Max Velocity	Max Relative Change
Extremely coarse	292.926 K	295.91 K	48.43%	0.81372 m s ⁻¹	—
Extra coarse	292.927 K	295.93 K	48.40%	0.80546 m s ⁻¹	1.02%
Coarser	292.955 K	295.97 K	48.44%	0.79959 m s ⁻¹	0.73%



Analytical approximation

Assumptions & simplifications:

The lettuce was considered a perfect sphere.

The surface temperature of the lettuce was uniform and equal to the ambient temperature.

Heat transfer was assumed to occur only in the radial direction.

Analytical approximation

Geometric reduction

$$D_I = 0.065 \text{ m},$$

$$R_I = \frac{D_I}{2} = 0.0325 \text{ m},$$

$$T_I = T_I(r, t), \quad 0 \leq r \leq R_I.$$

Transient radial conduction

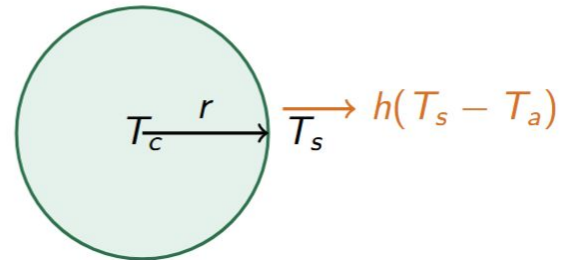
$$\rho_I C_{p,I} \frac{\partial T_I}{\partial t} = k_I \left[\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial T_I}{\partial r} \right) \right]$$

Conditions

$$T_I(r, 0) = T_{I,0},$$

$$\left. \frac{\partial T_I}{\partial r} \right|_{r=0} = 0,$$

$$-k_I \left. \frac{\partial T_I}{\partial r} \right|_{r=R_I} = h(T_s - T_a).$$



Analytical approximation

Estimating the Convective Coefficient from Nusselt Number

Dimensionless groups

$$Re = \frac{\rho_a U D_l}{\mu_a},$$

$$Pr = \frac{\mu_a C_{p,a}}{k_a},$$

$$Nu = \frac{h D_l}{k_a}.$$

Forced convection around a sphere

$$Nu = 2 + \left(0.4 Re^{1/2} + 0.06 Re^{2/3} \right) Pr^{0.4}$$

Heat-transfer coefficient and Biot number

$$h = \frac{Nu k_a}{D_l},$$

$$Bi = \frac{h R_l}{k_l}.$$

Analytical approximation

Define the temperature difference relative to surrounding air:

$$\theta(r, t) = T_I(r, t) - T_a, \quad \alpha_I = \frac{k_I}{\rho_I C_{p,I}}, \quad Fo = \frac{\alpha_I t}{R_I^2}.$$

Full transient spherical solution

$$\frac{\theta(r, t)}{\theta_i} = \sum_{n=1}^{\infty} A_n \frac{\sin(\lambda_n r / R_I)}{r / R_I} \exp(-\lambda_n^2 Fo)$$

Eigenvalue equation*

$$1 - \lambda_n \cot(\lambda_n) = Bi$$

Expansion coefficient

$$A_n = \frac{4[\sin(\lambda_n) - \lambda_n \cos(\lambda_n)]}{2\lambda_n - \sin(2\lambda_n)}$$

Analytical approximation

General temperature distribution

$$T_I(r, t) = T_a + (T_{I,0} - T_a) \sum_{n=1}^{\infty} A_n \frac{\sin(\lambda_n r / R_I)}{r / R_I} e^{-\lambda_n^2 Fo}$$

Center limit, $r \rightarrow 0$

$$T_I(0, t) = T_a + (T_{I,0} - T_a) \sum_{n=1}^{\infty} A_n \lambda_n e^{-\lambda_n^2 Fo}$$

Surface, $r = R_I$

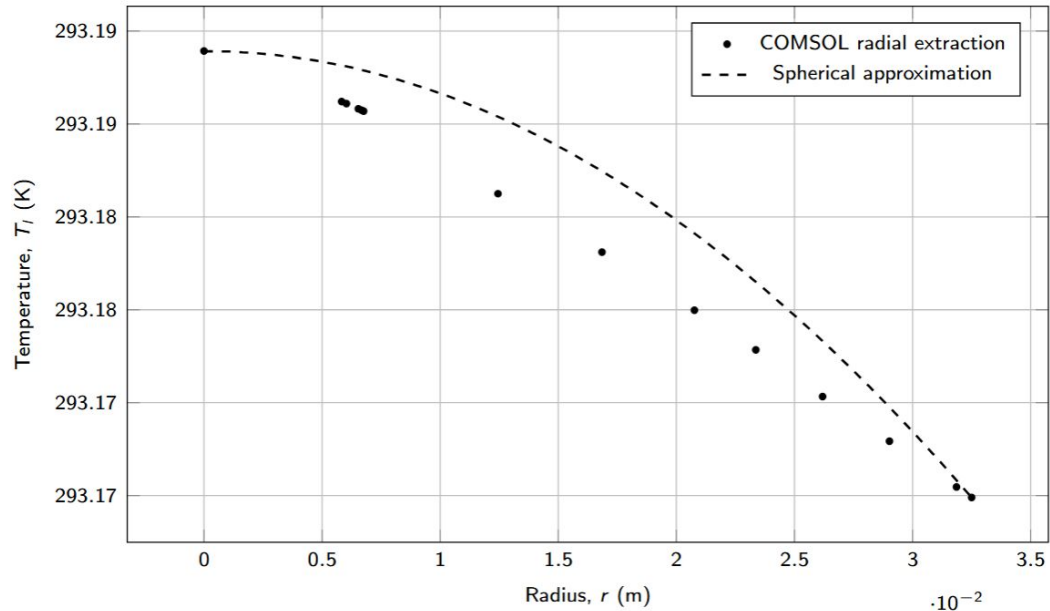
$$T_I(R_I, t) = T_a + (T_{I,0} - T_a) \sum_{n=1}^{\infty} A_n \sin(\lambda_n) e^{-\lambda_n^2 Fo}$$

Analytical approximation

Evaluation at $t = 180$ min

Surface-to-air temperature difference

$$\begin{aligned}T_a &= 293.15 \text{ K}, \\T_s - T_a &= 0.0149 \text{ K}, \\ \left. \frac{\partial T_l}{\partial r} \right|_{R_l} &= -\frac{h}{0.55} (0.0149).\end{aligned}$$



Parametric Analysis

Parameter ranges

- $v_{\text{in}} = 0.7, 0.8, 0.9, 1.1 \text{ m s}^{-1}$
- $T_{\text{LED}} = 313.15, 343.15, 383.15 \text{ K}$
- $T_{\text{in}} = 291.15, 293.15, 295.15 \text{ K}$
- Lettuce heat rate: $Q = 0, 0.004 \text{ W}$
- Initial/inlet RH: 65%

PID inlet-temperature regulation

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau + K_d \frac{de(t)}{dt}$$

$$e(t) = c_{\text{set}} - c(t),$$

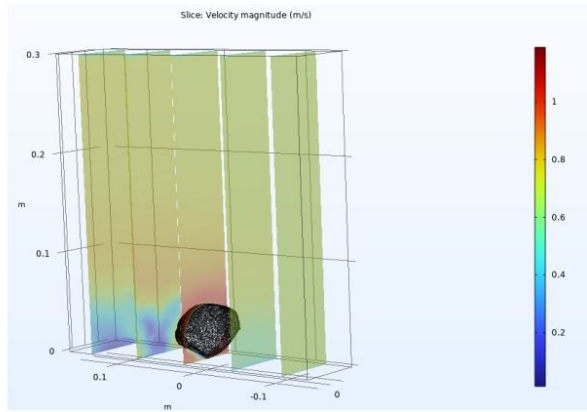
$$c_{\text{set}} = 293.15 \text{ K},$$

$$K_p = 0.5 \text{ or } 0.7, \quad K_i = 0.1, \quad K_d = 0.3.$$

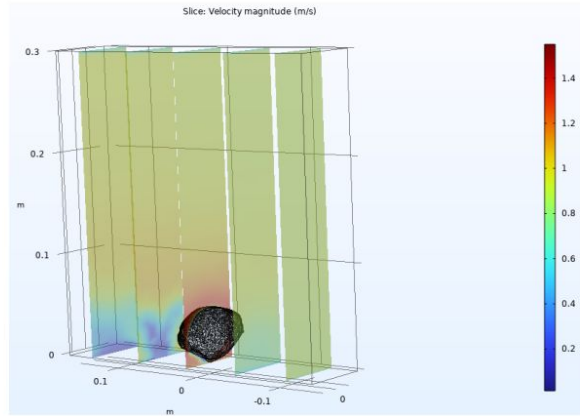
Parametric Analysis

Justification of V_{in} selection

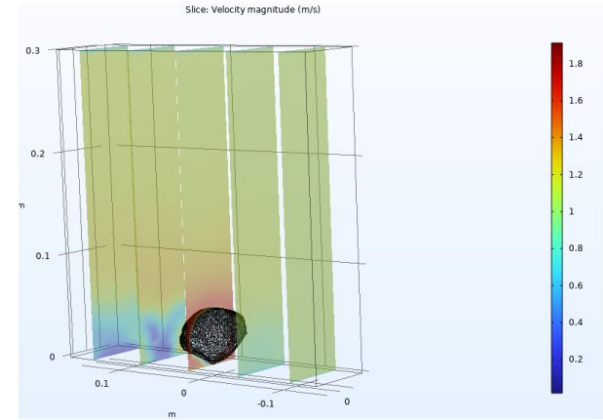
Optimal V at lettuce surface should be around $0.8 - 1.2 \text{ m/s}$



(a) $v_{in} = 0.7 \text{ m s}^{-1}$

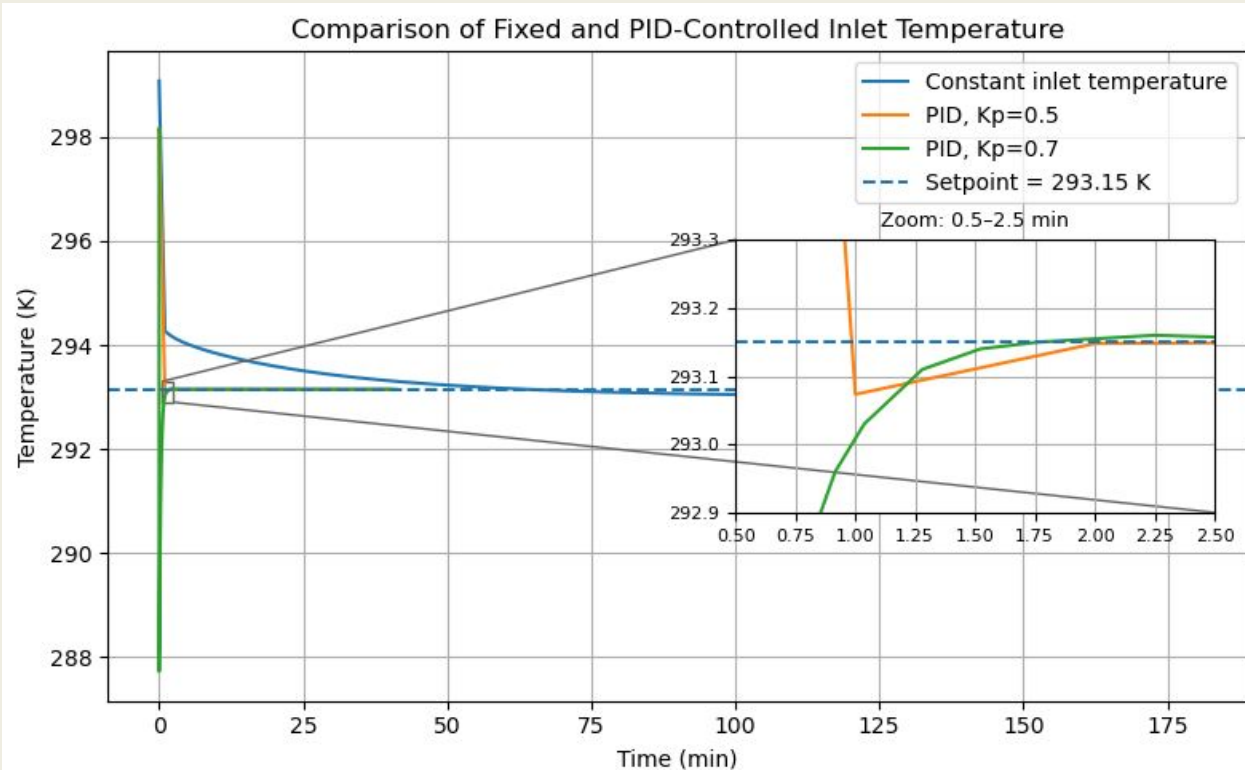


(b) $v_{in} = 0.9 \text{ m s}^{-1}$



(c) $v_{in} = 1.1 \text{ m s}^{-1}$

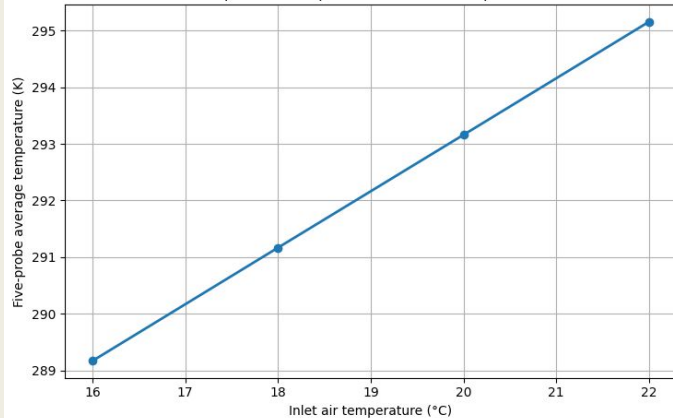
Parametric Analysis



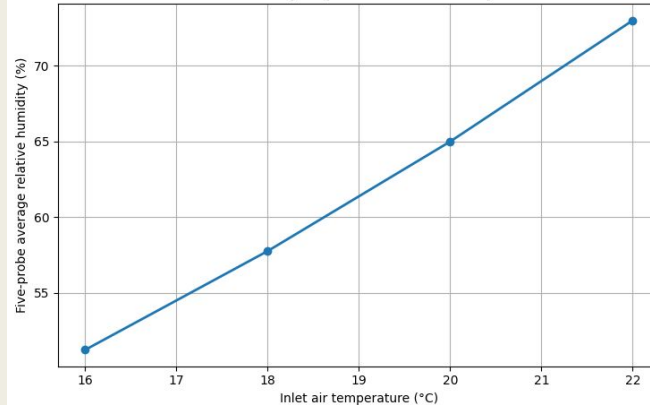
Sensitivity Analysis

Input Factor	Symbol	Levels Tested
Lettuce metabolic heat rate	Q	0, 0.004 W
LED boundary temperature	T_{LED}	40, 70, 110 °C
Inlet air temperature	T_{in}	16, 18, 20, 22 °C

Temperature response to inlet air temperature



Relative humidity response to inlet air temperature



Sensitivity Analysis

Sensitivity metrics

$$\Delta_i = \max_{x_i} (\mathbb{E}[Y|X_i = x_i]) - \min_{x_i} (\mathbb{E}[Y|X_i = x_i])$$

$$S_i = \frac{\text{Var}_{X_i} [\mathbb{E}(Y|X_i)]}{\text{Var}(Y)},$$

$$S_{T_i} = 1 - \frac{\text{Var}_{X_{\sim i}} [\mathbb{E}(Y|X_{\sim i})]}{\text{Var}(Y)}.$$

Table 11: Sobol sensitivity indices for average lettuce-surface temperature at $t = 180$ min.

Rank	Input Factor	First-Order Index S_i	Total-Order Index S_{T_i}
1	T_{in}	0.999999	0.999999
2	Q	1.09×10^{-6}	1.14×10^{-6}
3	T_{LED}	3.63×10^{-8}	2.01×10^{-7}

Table 12: Sobol sensitivity indices for average lettuce-surface relative humidity at $t = 180$ min.

Rank	Input Factor	First-Order Index S_i	Total-Order Index S_{T_i}
1	T_{in}	0.999999999	0.999999999
2	Q	3.42×10^{-10}	6.84×10^{-10}
3	T_{LED}	0.000000000	0.000000000

Discussion

Physical interpretation

- Lettuce geometry modifies the local flow field.
- Surface microclimate differs from boundary setpoints.
- PID control improves local temperature regulation.
- Analytical sphere model confirms reasonable radial thermal behavior.

Conclusion

- A coupled COMSOL model was developed for airflow, heat transfer, and water-vapor transport around scanned lettuce geometry.
- The spherical analytical approximation provided a useful benchmark for internal lettuce thermal response.
- Lettuce-surface velocity should be evaluated directly; $v_{in} = 0.8 \text{ m s}^{-1}$ was selected as the practical baseline.
- PID inlet-temperature control maintained the surface probe near 293.15 K more effectively than a fixed inlet boundary.
- Sensitivity results showed that T_{in} dominated lettuce-surface temperature and RH.